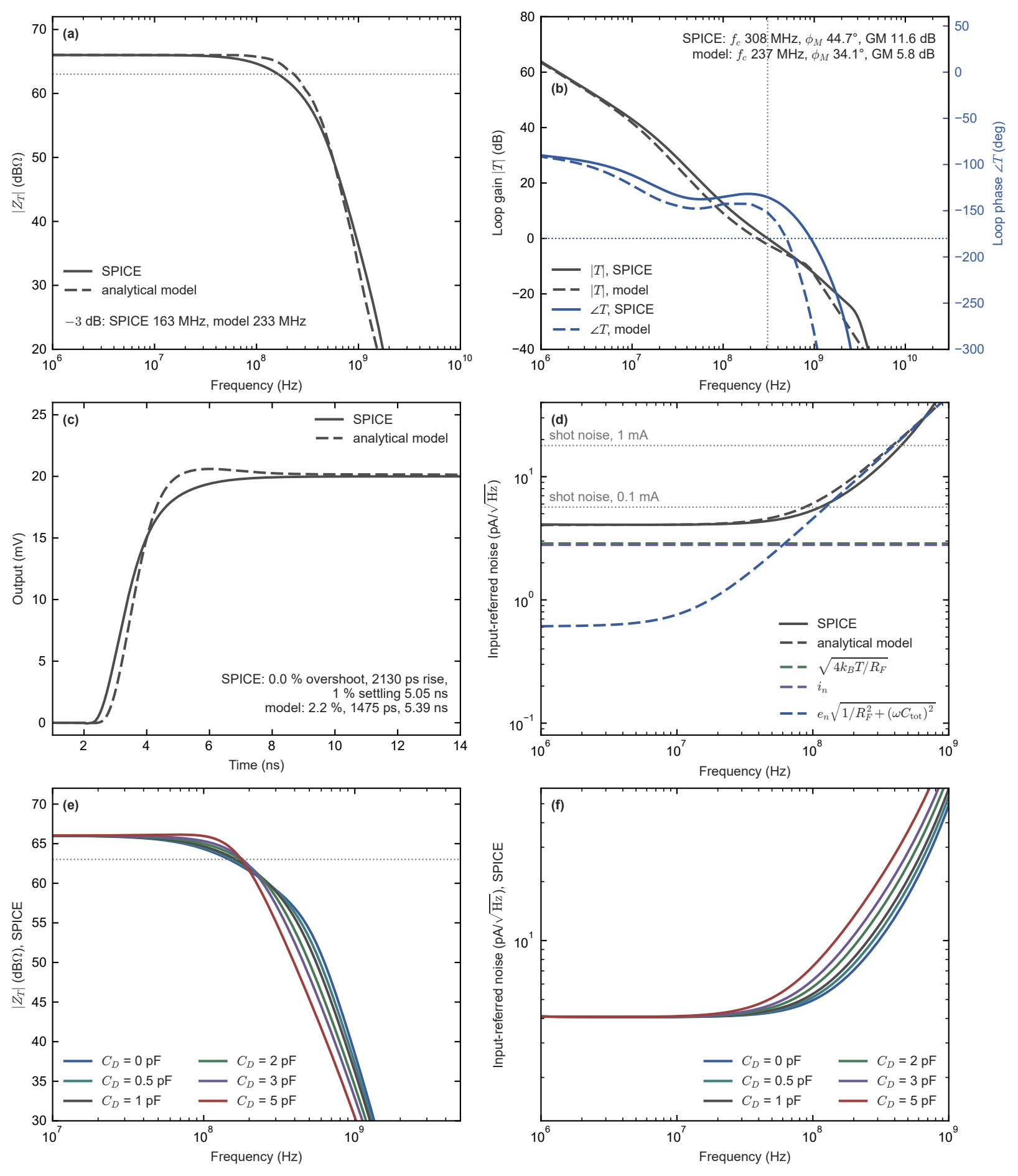

(a)
SPICE
analytical model
−3 dB: SPICE 163 MHz, model 233 MHz
|Z_T| (dBΩ)
Frequency (Hz)
(b)
SPICE: f_c 308 MHz, φ_M 44.7°, GM 11.6 dB
model: f_c 237 MHz, φ_M 34.1°, GM 5.8 dB
|T|, SPICE
|T|, model
∠T, SPICE
∠T, model
Loop gain |T| (dB)
Loop phase ∠T (deg)
Frequency (Hz)
(c)
SPICE
analytical model
SPICE: 0.0 % overshoot, 2130 ps rise,
1 % settling 5.05 ns
model: 2.2 %, 1475 ps, 5.39 ns
Output (mV)
Time (ns)
(d)
shot noise, 1 mA
shot noise, 0.1 mA
SPICE
analytical model
√(4k_BT/R_F)
i_n
e_n √(1/R_F² + (ωC_tot)²)
Input-referred noise (pA/√Hz)
Frequency (Hz)
(e)
C_D = 0 pF
C_D = 0.5 pF
C_D = 1 pF
C_D = 2 pF
C_D = 3 pF
C_D = 5 pF
|Z_T| (dBΩ), SPICE
Frequency (Hz)
(f)
C_D = 0 pF
C_D = 0.5 pF
C_D = 1 pF
C_D = 2 pF
C_D = 3 pF
C_D = 5 pF
Input-referred noise (pA/√Hz), SPICE
Frequency (Hz)